

Kai-Hung Huang

Sr. SOFTWARE ENGINEER AT XPERI (DTS) · ANDROID EXPERT

5F., No. 65, Ln. 182, Beixin Rd., Tamsui Dist., New Taipei City 251608, Taiwan

☎ (+886) 0925-133902 | ✉ randy.hunang@gmail.com | 🌐 www.github.com/Randy-Huang | 🔗 www.linkedin.com/in/kai-hung-huang-344117106/

Summary

OBJECTIVE

Seeking an Android software development, especially in Framework/API/Application layers, where I specialized in Java/C/C++ and 7+ work on AOSP.

ABOUT

- 7+ years experienced embedded Android/Linux system software (Android Framework/API/Application layer).
- 4+ years specialized in Android Framework (JNI/SystemService/Bluetooth) and customized SEpolicy.
- 2+ years supervised a team to efficiently customize SurfaceFlinger/API/Settings for integrating features to 3rd developer.
- 2+ years analyzed Android system performance about stability and CPU/memory usage using systrace and Android profiler.
- ~1 year worked on sensor domain of Android BSP development.
- A passionate person who focuses on programming and has good communication skills, and excellent team player.

Work Experience

XPERI Corporation (DTS)

Taipei, Taiwan

Sr. SOFTWARE ENGINEER

Aug. 2021 - Present

- Developed list-style layout using the extended RecyclerView with GridLayoutManager and customized Adapter for optimized UI representation.
- Maintained streaming techniques with RTP/RTCP network protocols transmitting audio data for DTS Play-Fi wireless multi channel.
- Customized a linux process which can receive audio PCM data from system using unix domain socket based on Android TV.

XRSPACE Inc.

Taipei, Taiwan

LEAD PROGRAMMER

Mar. 2019 - Aug. 2021

- Lead team members (4 people) to reach the project tasks.
- Coordinated and host task topics for cross-team collaboration to decrease unnecessary effort.
- Developed a library that connects between HMD and mobile device using BT module, which customized encryption/decryption and packet.
- Customized SurfaceView/SurfaceFlinger to cast specified layers of VR content from a foreground VR application to TV by Miracast.
- Implemented a library dedicating specified CPU to a particular process or function by using the technique of CPU affinity.
- Experienced with analyzing the system performance and stability using cmd tools/systrace/Android profiler.
- Experienced with fine-tuning an application performance for UI rendering with systrace.

Sr. SOFTWARE ENGINEER

Sep. 2018 - Mar. 2019

- Developed camera (mono, RGB, depth) getting from QVR (Qualcomm VR solution) and Android Camera NDK for VR 6-Dof processing.
- Implemented a Ring Buffer data structure to store camera frame and transmit it to clients (Java/Unity layers) through ashmem or fd.
- Designed a preview window tool by ANativeWindow to display camera frame using YUV-420/10-bit Raw RGB format.
- Designed communication architecture between VR headset and Android phone using Bluetooth LE.

ASUSTeK Computer Inc.

Taipei, Taiwan

Sr. SOFTWARE ENGINEER

Dec. 2015 - Sep. 2018

- [Zenbo Junior] Developed the behavior of "Line Following" feature with PID control algorithm.
- [Zenbo] Developed RobotAPI and RobotSDK for 3rd Developer based on Android Framework.
- [Zenbo] Developed the behavior of moving to Docking charger and integrate MCU command for charger control.
- [Zenbo] Integrated Battery/Facial/Motion features of main functions using Android Service to RobotFramework.
- [Zenbo] Android BSP development - integrate motion sensors (accelerometer, gyroscope, and compass) using Intel ISH (sensor hub).
- [Zenbo] Android BSP development - porting capacitive touch sensor (GPIO, BIOS, Linux driver, Android HAL/Framework).
- Android App unit test tool for sensor testing/debugging.

Education

TKU(Tamkang University)

New Taipei City, Taiwan

MASTER OF ENGINEERING, ROBOTICS ENGINEERING

Mar. 2013 - Aug. 2014

- Grade: GPA: 4 and Course Average: 91.36/100.
- Graduated in one academic year (Excellent MS student qualification)
- Scholarships (over \$100,000+): Awarded outstanding academic achievement for 2 semesters

BACHELOR OF ENGINEERING, ELECTRICAL ENGINEERING

Mar. 2009 - Aug. 2013

- Grade: GPA: 4 and Course Average: 92.60/100.
- Academic Honors: Graduated in the honor of the top-rated prize, and Awarded 1st excellent academic for 7 semesters.
- Scholarships (over \$100,000+): Awarded excellent academic for 8 consecutive semesters.